

Table 8: Statistical significance analysis on representative long-term forecasting settings with  $H = 96$ . We report mean and standard deviation over three independent runs with different random seeds.  $p$ -values are computed using a paired two-sided  $t$ -test between RDTU and Time-LLM over matched dataset-seed pairs.

| Dataset     | Metric    | RDTU                                  | Time-LLM                              | Relative Gain  | $p$ -value    |
|-------------|-----------|---------------------------------------|---------------------------------------|----------------|---------------|
| ETTh1       | MSE / MAE | $0.352 \pm 0.003$ / $0.383 \pm 0.002$ | $0.376 \pm 0.004$ / $0.403 \pm 0.003$ | 6.38% / 4.96%  | 0.018 / 0.021 |
| Electricity | MSE / MAE | $0.125 \pm 0.002$ / $0.211 \pm 0.002$ | $0.137 \pm 0.002$ / $0.233 \pm 0.003$ | 8.76% / 9.44%  | 0.014 / 0.012 |
| Traffic     | MSE / MAE | $0.359 \pm 0.004$ / $0.236 \pm 0.003$ | $0.393 \pm 0.005$ / $0.268 \pm 0.004$ | 8.65% / 11.94% | 0.011 / 0.009 |
| Average     | MSE / MAE | $0.279 \pm 0.003$ / $0.277 \pm 0.002$ | $0.302 \pm 0.004$ / $0.301 \pm 0.003$ | 7.62% / 7.97%  | 0.013 / 0.015 |